

## **BACKLOG BUDDY ? STUDY MATERIAL**

Electromagnetic Theory and Transmission Lines ? Unit-wise Lecture Notes

Subject Code: EC404 Semester: 4

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Condensed notes covering all five units: Electrostatics, Magnetostatics, Maxwell's Equations, Electromagnetic Wave Propagation, Transmission Lines.

### **Unit 1: Electrostatics**

- Definition, key concepts and significance of Electrostatics
- Important diagrams / derivations / algorithms for Electrostatics
- Solved examples and previous-year questions on Electrostatics

### **Unit 2: Magnetostatics**

- Definition, key concepts and significance of Magnetostatics
- Important diagrams / derivations / algorithms for Magnetostatics
- Solved examples and previous-year questions on Magnetostatics

### **Unit 3: Maxwell's Equations**

- Definition, key concepts and significance of Maxwell's Equations
- Important diagrams / derivations / algorithms for Maxwell's Equations
- Solved examples and previous-year questions on Maxwell's Equations

### **Unit 4: Electromagnetic Wave Propagation**

- Definition, key concepts and significance of Electromagnetic Wave Propagation
- Important diagrams / derivations / algorithms for Electromagnetic Wave Propagation
- Solved examples and previous-year questions on Electromagnetic Wave Propagation

### **Unit 5: Transmission Lines**

- Definition, key concepts and significance of Transmission Lines
- Important diagrams / derivations / algorithms for Transmission Lines
- Solved examples and previous-year questions on Transmission Lines

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- \* Revise definitions, diagrams and standard derivations ? they earn easy marks.
- \* Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

**Disclaimer: Sample study material for preparation support only.**

Appearance counts are based on available papers and are not exam predictions.